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LINEAR ALGEBRA

Sem 251 - CC02 - Group 02

**PRINCIPAL COMPONENT ANALYSIS: DATA DIMENSIONALITY
REDUCTION**

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INTRODUCTION

In the era of big data, researchers and analysts are frequently confronted with datasets containing a vast number of variables. While high-dimensional data provides a wealth of information, it introduces the "Curse of Dimensionality," leading to computational inefficiency, overfitting in machine learning models, and difficulties in visualization. Principal Component Analysis (PCA) stands as a cornerstone technique in statistical data analysis to address these challenges. By employing orthogonal linear transformations, PCA projects high-dimensional data onto a lower-dimensional subspace while maximizing the preservation of variance[cite: 28, 604].

This report explores the theoretical underpinnings and practical applications of PCA. We examine the intersection of statistics and linear algebra—specifically how concepts such as covariance matrices, eigenvectors, and eigenvalues are synthesized to identify latent structures within complex datasets. Furthermore, this project bridges theory and practice by applying PCA to a real-world dataset: the history of natural disasters in Vietnam.

The primary objectives of this project are:

- **Theoretical Mastery:** To elucidate the mathematical foundations of PCA, focusing on eigen-decomposition and singular value decomposition (SVD).
- **Algorithm Implementation:** To demonstrate the step-by-step mechanism of the PCA algorithm, including data standardization and feature vector selection.
- **Practical Application:** To utilize Python and the `scikit-learn` library to analyze the Vietnam Natural Disaster dataset, reducing its dimensionality to reveal correlations between economic damage and human casualties.

Chapter 1

OVERVIEW

1.1 Problem Definition: Data Dimensionality

1.1.1 Definition of Data Dimensionality

Data dimensionality refers to the number of features (variables or attributes) present in a dataset. For a dataset with n samples and p features, the dimensionality is defined as p .

1.1.2 The Curse of Dimensionality

High-dimensional data presents several challenges, often referred to as the "Curse of Dimensionality". These challenges include:

- **Exponential increase in volume:** As dimensions increase, the data becomes sparse.
- **Distance metrics:** Distances between data points become less meaningful in high-dimensional space.
- **Computational cost:** High dimensionality leads to increased computational cost and memory usage.
- **Overfitting:** There is a higher risk of overfitting in machine learning models.
- **Visualization:** It is difficult for humans to visualize data beyond 3 dimensions.

1.1.3 Why Dimensionality Reduction is Needed

Dimensionality reduction is crucial to remove redundant or noisy features and retain the most important information. It helps to speed up algorithms, improve model generalization, and enable better visualization of complex datasets.

1.1.4 Types of Dimensionality Reduction Techniques

Techniques are generally categorized into:

- **Linear methods:** Such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA).
- **Non-linear methods:** Such as t-SNE, UMAP, and Autoencoders.
- **Justification for choosing PCA:** For this study, we selected PCA over non-linear methods like t-SNE because **interpretability** is our priority. While t-SNE is excellent for clustering, it distorts the feature space, making it impossible to tell which original variables (e.g., "Deaths" vs "Damage") caused the separation. PCA preserves the linear weights (loadings), allowing us to construct a Biplot that explicitly links disaster severity to specific economic or human costs.

1.2 Introduction to PCA

1.2.1 What is Principal Component Analysis?

Principal Component Analysis (PCA) is a linear transformation technique that projects high-dimensional data onto a new coordinate system known as principal components. In this new system:

- The **first principal component** captures the direction of maximum variance.
- Each **subsequent component** captures the maximum remaining variance while being orthogonal to the previous ones.

1.2.2 Historical Background

PCA was invented by Karl Pearson in 1901 and later developed by Harold Hotelling in 1933 as a statistical tool.

1.2.3 Key Intuition

The key intuition behind PCA is to find the "best" lower-dimensional subspace that preserves as much variability (information) as possible from the original dataset.

1.2.4 Illustrative Example

Consider a 2D dataset, such as the height versus weight of people. PCA rotates the axes to align with the direction of the greatest spread (variance) of the data.

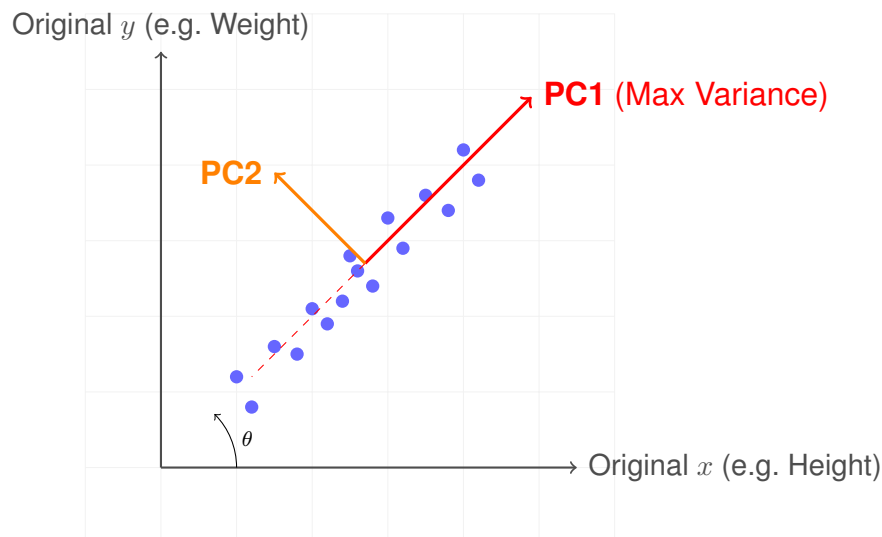


Figure 1.1: Visualizing PCA: The red axes (PC_1, PC_2) are rotated to align with the data's greatest spread, distinct from the original x, y axes.

1.2.5 Relationship with Other Methods

PCA is closely related to Singular Value Decomposition (SVD) and can be viewed as a special case of factor analysis.

1.3 Objectives, Advantages, and Disadvantages

1.3.1 Main Objectives

The primary goals of applying PCA are:

- Maximize preserved variance while minimizing the number of dimensions.
- Decorrelate features (principal components are orthogonal and uncorrelated).
- Enable visualization of high-dimensional data (e.g., reducing to 2D or 3D).
- Improve performance and reduce overfitting in downstream machine learning tasks.
- Reduce noise by discarding low-variance components.

1.3.2 Advantages

PCA offers several significant benefits:

- **Efficiency:** It is computationally efficient and scalable, especially with SVD-based implementations.

- **Unsupervised:** It does not require labeled data.
- **Deterministic:** The same input always yields the same result.
- **Accessibility:** It is widely available in libraries such as scikit-learn, MATLAB, and R.
- **Foundation:** It has a strong mathematical foundation and interpretability regarding variance.

1.3.3 Disadvantages

Despite its utility, PCA has limitations:

- **Linearity Assumption:** PCA assumes linear relationships and fails to capture non-linear structures (manifold learning or Kernel PCA may be required).
- **Interpretability:** Principal components are linear combinations of original features, which can be semantically hard to interpret.
- **Scaling Sensitivity:** Features must be standardized beforehand as PCA is sensitive to the scale of data.
- **Outliers:** PCA is sensitive to outliers, which can dominate the variance.
- **Information Definition:** It assumes variance is the best measure of information, which is not always true for classification tasks.

Chapter 2

Mathematical Foundations

This chapter establishes the statistical and algebraic framework necessary for understanding Principal Component Analysis (PCA), including expectation, covariance, and eigen-decomposition.

2.1 Statistical Features

2.1.1 Expectation

The expectation of a random variable X , denoted $E[X]$, is the theoretical long-run average value of X . It represents the center or "balance point" of the distribution.

For a Discrete Random Variable: If X takes values $x_1, x_2, \dots, x_n$ with probabilities $p_1, p_2, \dots, p_n$:

$$E[X] = \sum_{i=1}^n x_i p_i \quad (2.1)$$

For a Continuous Random Variable: If X has a probability density function $f(x)$:

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx \quad (2.2)$$

2.1.2 Variance

Variance measures how spread out the values of a random variable are around the mean. A higher variance means the data is more dispersed.

Formula:

$$\text{Var}(X) = E[(X - E[X])^2] \quad (2.3)$$

This is the expected (average) squared distance from the mean.

Equivalent Computational Formula:

$$\text{Var}(X) = E[X^2] - (E[X])^2 \quad (2.4)$$

2.1.3 Standard Deviation

Standard deviation is the square root of variance:

$$\sigma = \sqrt{\text{Var}(X)} \quad (2.5)$$

2.2 Covariance

2.2.1 Definition (Scalar)

For integrable random variables X and Y :

$$\text{Cov}(X, Y) := E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y] \quad (2.6)$$

Note that the variance of X is simply the covariance of X with itself:

$$\text{Var}(X) = \text{Cov}(X, X)$$

2.2.2 Properties

- **Symmetry:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.

- **Bilinearity:** Covariance is bilinear in both arguments:

$$\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)$$

- **Relation to Independence:** If X and Y are independent (and integrable), then $\text{Cov}(X, Y) = 0$.

– *Note:* The converse is false; covariance zero does **not** imply independence in general.

- **Cauchy–Schwarz Bound:**

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)\text{Var}(Y)}$$

This yields the **correlation coefficient** $\rho_{X,Y}$:

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}, \quad |\rho| \leq 1$$

2.2.3 Covariance Matrix (Multivariate)

Let $X = (X_1, \dots, X_n)^\top$ be a random vector with finite second moments. Define the **covariance matrix** Σ :

$$\Sigma := \text{Cov}(X) = E[(X - \mu)(X - \mu)^\top], \quad \text{where } \mu = E[X] \quad (2.7)$$

The entries are given by $\Sigma_{ij} = \text{Cov}(X_i, X_j)$.

Key Facts:

- Σ is symmetric: $\Sigma^\top = \Sigma$.
- Σ is positive semidefinite (PSD): For any vector $a \in \mathbb{R}^n$:

$$a^\top \Sigma a = \text{Var}(a^\top X) \geq 0$$

- If all components have finite variance, then Σ is finite.

2.2.4 Conditional Covariance and Law of Total Covariance

- **Law of Total Expectation:** $E[X] = E[E[X|Y]]$.
- **Law of Total Covariance:**

$$\text{Cov}(X, Z) = E[\text{Cov}(X, Z|Y)] + \text{Cov}(E[X|Y], E[Z|Y]) \quad (2.8)$$

- **Law of Total Variance** (Special case):

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

2.2.5 Whitening / Decorrelation

Given a positive definite covariance matrix Σ , one can compute a **whitening matrix** W such that:

$$W\Sigma W^\top = I$$

Common choices for W :

- $W = \Sigma^{-1/2}$ (via eigen-decomposition).
- $W = L^{-1}$ (via Cholesky decomposition $\Sigma = LL^\top$).

Whitening removes covariances (resulting in an identity covariance matrix) and is often used as preprocessing for algorithms like ICA.

2.2.6 Difference Between Diagonal and Off-Diagonal Values

A covariance matrix summarizes how pairs of features vary together. For features $X_1, \dots, X_n$:

$$\Sigma = \begin{pmatrix} \text{Cov}(X_1, X_1) & \cdots & \text{Cov}(X_1, X_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \cdots & \text{Cov}(X_n, X_n) \end{pmatrix}$$

1. Diagonal Values (Variances): Entries $\text{Cov}(X_i, X_i) = \text{Var}(X_i)$ describe the spread of a single dimension.

- **Large variance:** The feature values vary significantly.

- **Small variance:** The feature values are tightly clustered around the mean.

2. Off-Diagonal Values (Covariances): Entries $\text{Cov}(X_i, X_j)$ where $i \neq j$ describe how two features vary together.

- **Positive:** When X_i increases, X_j tends to increase.
- **Negative:** When X_i increases, X_j tends to decrease.
- **Zero:** Features are uncorrelated.

These values reflect the shape and orientation of the data cloud in multivariate space.

2.3 Eigenvalues and Eigenvectors

2.3.1 Algebraic Definition

For a square $n \times n$ matrix A , a scalar λ is an **eigenvalue** and a non-zero vector v is an **eigenvector** if:

$$Av = \lambda v \quad (2.9)$$

Equivalently, λ is a root of the characteristic polynomial:

$$p_A(\lambda) := \det(A - \lambda I) = 0$$

Multiplicity:

- **Algebraic multiplicity:** Multiplicity of λ as a root of $p_A(\lambda)$.
- **Geometric multiplicity:** Dimension of the nullspace $\ker(A - \lambda I)$ (number of linearly independent eigenvectors).
- Always $1 \leq \text{geom mult} \leq \text{alg mult}$.

2.3.2 Spectral Properties

- **Trace:** $\text{tr}(A) = \sum \lambda_i$ (sum of eigenvalues).
- **Determinant:** $\det(A) = \prod \lambda_i$ (product of eigenvalues).
- **Real Symmetric Matrices:** All eigenvalues are real, and eigenvectors can be chosen to be orthonormal (Spectral Theorem).

2.3.3 Singular Value Decomposition (SVD)

For any $m \times n$ real matrix M , SVD writes:

$$M = U\Sigma V^T \quad (2.10)$$

Where U and V are orthogonal matrices, and Σ is a diagonal matrix with non-negative **singular values** $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

- The singular values of M are the square roots of the eigenvalues of $M^T M$ (or MM^T).
- SVD generalizes eigen-decomposition to non-square matrices and is numerically stable.

2.3.4 Eigen-decomposition of Covariance Matrices & PCA

The covariance matrix Σ is symmetric and positive semidefinite. Its eigen-decomposition is:

$$\Sigma = Q\Lambda Q^T \quad (2.11)$$

where Q is orthonormal and $\Lambda = \text{diag}(\lambda_1 \geq \lambda_2 \geq \dots \geq 0)$.

Relation to PCA:

- **Principal Components:** The eigenvectors corresponding to the largest eigenvalues.
- **Variance Explained:** The variance explained by the k -th principal component is equal to λ_k .
- PCA projects mean-centered data onto the first r eigenvectors to create a lower-dimensional representation that preserves maximal variance.

Chapter 3

The Principal Component Analysis Algorithm

3.1 Theoretical Framework

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a high-dimensional dataset into a smaller set of uncorrelated variables called Principal Components, while retaining as much of the original variation as possible [1]. The algorithm follows a rigorous five-step process.

3.1.1 Step 1: Standardization (Data Scaling)

The first step is to standardize the range of the continuous initial variables so that each of them contributes equally to the analysis. If there are large differences between the ranges of initial variables, those variables with larger ranges will dominate over those with small ranges.

For example, a "Damage Cost" variable ranging in the millions will mathematically overpower a "Death Count" variable ranging in the tens [2]. Transforming the data to comparable scales prevents this problem. We typically do this using the Z-score normalization:

$$z = \frac{x - \mu}{\sigma} \quad (3.1)$$

Where x is the original value, μ is the mean, and σ is the standard deviation. This results in a dataset where every variable has a mean of 0 and a variance of 1 [2].

3.1.2 Step 2: Covariance Matrix Computation

The aim of this step is to understand how the variables of the input data set are varying from the mean with respect to each other. The covariance matrix (Σ) is a symmetric matrix that has as entries the covariances associated with all possible pairs of the initial variables [3].

$$\Sigma = \frac{1}{n-1} X^T X \quad (3.2)$$

Since the covariance of a variable with itself is its variance, the main diagonal contains the variances of each initial variable. High positive values indicate variables that increase together, while zero values indicate no relationship.

3.1.3 Step 3: Compute the Eigenvectors and Eigenvalues

Eigenvectors and eigenvalues are the linear algebra concepts that we need to compute from the covariance matrix in order to determine the principal components of the data [3].

- **Eigenvectors** (v): Determine the directions of the new feature space (where the data is most spread out).
- **Eigenvalues** (λ): Determine the magnitude (or variance) of the data along those new directions.

Mathematically, we solve the characteristic equation:

$$\det(\Sigma - \lambda I) = 0 \quad (3.3)$$

3.1.4 Step 4: Feature Vector Selection

Once eigenvectors are found, we order them by their eigenvalues, from highest to lowest ($\lambda_1 \geq \lambda_2 \dots$). This gives us the components in order of significance. The eigenvector with the highest eigenvalue is the First Principal Component (PC1) [1].

We then choose to keep the top k eigenvectors to form our feature vector W . This is the step where dimensionality reduction actually occurs, as we discard the components with low information (noise).

3.1.5 Step 5: Projection (Recasting the Data)

In the final step, we use the feature vector W formed in the previous step to reorient the data from the original axes to the ones represented by the principal components. This is done by matrix multiplication [3]:

$$Z = X \cdot W \quad (3.4)$$

The resulting matrix Z represents the original data compressed into fewer dimensions.

3.2 Illustrative Example: Manual Verification

To demonstrate the mathematical mechanics of the algorithm described above, we perform a manual calculation on a simplified dataset.

Problem: We have data for 3 students involving their study hours (Variable x) and their test scores (Variable y). We wish to reduce this 2D data into 1D using PCA.

The Dataset:

Table 3.1: Dataset for Manual Calculation Example

Student	Study Hours (x)	Test Score (y)
A	2	8
B	4	12
C	6	16

Execution

1. Standardization (Centering):

First, we calculate the means: $\bar{x} = 4$, $\bar{y} = 12$. We center the data by subtracting the means from each point:

$$X_{cen} = \begin{pmatrix} 2-4 & 8-12 \\ 4-4 & 12-12 \\ 6-4 & 16-12 \end{pmatrix} = \begin{pmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{pmatrix}$$

2. Covariance Matrix:

We calculate the covariance matrix Σ using the formula $\Sigma = \frac{1}{n-1} X^T X$. Since $n = 3$, we divide by 2.

$$\Sigma = \frac{1}{2} \begin{pmatrix} -2 & 0 & 2 \\ -4 & 0 & 4 \end{pmatrix} \begin{pmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 8 \\ 8 & 16 \end{pmatrix}$$

3. Eigenvalues:

We solve the characteristic equation $\det(\Sigma - \lambda I) = 0$:

$$\det \begin{pmatrix} 4-\lambda & 8 \\ 8 & 16-\lambda \end{pmatrix} = 0$$

$$(4-\lambda)(16-\lambda) - 64 = 0 \Rightarrow \lambda^2 - 20\lambda = 0$$

Solving for λ gives two results:

- $\lambda_1 = 20$ (This is PC1, containing all the information).
- $\lambda_2 = 0$ (This is PC2, containing no information).

4. Eigenvectors:

We find the eigenvector for the largest eigenvalue ($\lambda_1 = 20$) by solving the system $(\Sigma - 20I)v = 0$:

$$\begin{pmatrix} -16 & 8 \\ 8 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This simplifies to: $-16x + 8y = 0 \Rightarrow 2x = y$. We choose a simple vector that fits this rule: $v = [1, 2]^T$.

To make it a unit vector (length of 1), we divide by $\sqrt{1^2 + 2^2} = \sqrt{5}$:

$$u_1 = \begin{pmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{pmatrix} \approx \begin{pmatrix} 0.45 \\ 0.89 \end{pmatrix}$$

5. Projection:

We project the centered data onto our new axis (u_1) to get the final reduced data (Z):

$$Z = X_{cen} \cdot u_1 = \begin{pmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 0.45 \\ 0.89 \end{pmatrix} = \begin{pmatrix} -4.47 \\ 0 \\ 4.47 \end{pmatrix}$$

Conclusion: We successfully reduced the data from 2 coordinates (Hours, Score) to just 1 coordinate (Z) while capturing 100% of the variance.

3.3 Verification on Real Dataset Subset

To further validate our approach, we perform the same manual calculation on a subset of the actual *Vietnam Natural Disaster* dataset. We selected three historical events from the 1960s where both human and economic impacts were recorded to observe how the large variance in "Damage" influences the principal components.

The Subset (Raw Data):

Table 3.2: Subset of Vietnam Disasters for Manual Verification

Event	Type	Deaths (x)	Damage ('000 \$) (y)
1	Typhoon Iris (1964)	7,000	471,770
2	Riverine Flood (1966)	31	90,163
3	Fire Accident (1967)	3	4,387

Step-by-Step Execution

1. Calculation of Means:

First, we compute the average for each feature:

$$\bar{x} = \frac{7000 + 31 + 3}{3} \approx 2,344.67$$

$$\bar{y} = \frac{471770 + 90163 + 4387}{3} \approx 188,773.33$$

2. Centering the Data ($X - \mu$):

We subtract the mean from each data point to center the dataset at (0,0):

$$X_{cen} = \begin{pmatrix} 4655.33 & 282996.67 \\ -2313.67 & -98610.33 \\ -2341.67 & -184386.33 \end{pmatrix}$$

3. Covariance Matrix (Σ):

We calculate the covariance matrix using $\Sigma = \frac{1}{n-1} X_{cen}^T X_{cen}$ (with $n = 3$):

$$\Sigma \approx \begin{pmatrix} 1.63 \times 10^7 & 9.89 \times 10^8 \\ 9.89 \times 10^8 & 6.19 \times 10^{10} \end{pmatrix}$$

Observation: The variance of Variable y (Damage) is approximately 6.19×10^{10} , which is significantly larger than the variance of x (Deaths). This mathematically confirms why standardization (Log Transformation/Scaling) is critical for this dataset; otherwise, "Damage" would completely dominate the analysis.

4. Eigenvalues and Eigenvectors:

Solving the characteristic equation $\det(\Sigma - \lambda I) = 0$ yields two eigenvalues:

- $\lambda_1 \approx 6.19 \times 10^{10}$ (Accounts for $> 99\%$ of the variance).

- $\lambda_2 \approx 4.64 \times 10^5$ (Accounts for $< 1\%$ of the variance).

The eigenvector corresponding to PC1 is approximately $v_1 = [-0.016, -0.999]^T$. The value -0.999 indicates that PC1 is almost entirely aligned with the "Damage" axis.

5. Projection:

Finally, we project the centered data onto the new PC1 axis ($Z = X_{cen} \cdot v_1$):

$$Z \approx \begin{pmatrix} -283,035 \\ 98,635 \\ 184,400 \end{pmatrix}$$

This step successfully reduces our 2-dimensional data (Deaths, Damage) into a 1-dimensional representation (Z) while preserving the vast majority of the variance found in the original subset.

Chapter 4

Applications of PCA

Principal Component Analysis is a versatile tool used across various domains to manage high-dimensional data. This chapter explores its critical applications in Image Processing, Finance, Biology, and Machine Learning.

4.1 Image Processing

4.1.1 Pattern Recognition

Images are inherently high-dimensional; for instance, a small 28×28 pixel image contains 784 features. PCA reduces these features to a small number (typically 40–60) while preserving the main structure. The principal components often represent fundamental shapes such as vertical strokes, curves, or loops. Machine learning models trained on these reduced images run faster, generalize better, and require less storage. This makes PCA a staple in character recognition tasks.

4.1.2 Background Subtraction in Videos

Videos often consist of static backgrounds (walls, roads) and moving foreground objects. PCA can identify the background because it appears in almost every frame, resulting in very high variance stability. Conversely, moving objects change rapidly and are captured in low-variance components. By subtracting the PCA-reconstructed background, systems can isolate moving objects for tracking, motion detection, and surveillance.

4.1.3 Image Denoising

Random noise in images is typically uncorrelated and appears in the low-variance components (the "tail" of the eigenvalue spectrum). By reconstructing an image using only the top principal components, we preserve edges and large structures while filtering out the fine-grain noise. This technique is effective in medical imaging and astronomy.

4.2 Finance

4.2.1 Yield Curve Analysis

In fixed income markets, the yield curve describes interest rates for bonds of different maturities. PCA reveals that complex curve movements are driven by three main components:

- **Level (PC1):** An overall up/down shift of interest rates.
- **Slope (PC2):** The difference between short-term and long-term rates.
- **Curvature (PC3):** How "bowed" the curve is.

Traders and central banks use these factors to model risk and understand economic expectations.

4.2.2 Stress Testing Portfolios

Regulatory frameworks require banks to test portfolios against extreme events. PCA identifies components representing major risk sources. By "shocking" these components (e.g., forcing PC1 to jump by 5 standard deviations), analysts can simulate realistic stress scenarios that capture how markets move together, providing more accurate risk estimates than shocking individual assets.

4.2.3 Identifying Correlated Assets

Markets contain assets that behave similarly due to hidden factors. PCA applied to return matrices reveals which assets cluster together. This helps investors build diversified portfolios and avoid accidental overexposure to specific hidden risks (e.g., a common factor driving all commodities).

4.3 Biology and Bioinformatics

4.3.1 Single-Cell RNA Sequencing (scRNA-seq)

Modern biology datasets are massive, often containing 20,000+ gene expression values per cell. PCA compresses this space into 30–50 components representing major cell states and biological processes. This allows researchers to cluster cells into distinct types (e.g., T-cells, Neurons) which would be impossible in the noisy, high-dimensional raw space.

4.3.2 Protein Structure Analysis

Proteins are dynamic molecules. Simulations of their movement generate high-dimensional atomic coordinate data. PCA identifies dominant motions, such as the opening/closing of

protein domains or twisting motions, helping researchers understand how proteins fold and interact with drugs.

4.4 Machine Learning

4.4.1 Noise Reduction

Real-world data from sensors or IoT devices often contains noise. Since noise usually contributes very little variance, it is relegated to the lowest principal components. By dropping these components, PCA acts as an automated filter, cleaning the input data and improving the robustness of downstream models.

4.4.2 Feature Engineering

Raw features often contain redundant information. PCA creates new composite features that capture the strongest signals and remove collinearity. For tasks like credit scoring or churn prediction, these uncorrelated features help algorithms learn more efficiently.

4.4.3 Speeding Up Clustering

Clustering algorithms like K-Means or DBSCAN struggle in high dimensions due to computational cost and the breakdown of distance metrics. Reducing thousands of features to a few dozen principal components makes clustering faster, more accurate, and more stable.

Chapter 5

Case Study: Analysis of Vietnam Natural Disasters

In this chapter, we apply Principal Component Analysis (PCA) to a real-world dataset to demonstrate its effectiveness in reducing dimensionality and identifying patterns in complex data.

5.1 Data Description

This analysis utilizes the *Vietnam Natural Disaster* dataset, which records historical disaster events from 1953 to 2023. To analyze the severity and dimensionality of these events, we selected four key numerical variables that measure impact:

- **Total Deaths:** The number of fatalities per event.
- **No. Injured:** The number of people injured.
- **Total Affected:** The total population affected (homeless, evacuated, etc.).
- **Total Damage:** The economic cost adjusted for inflation (in thousands of US Dollars).

Table 5.1: Sample of the Vietnam Natural Disaster Dataset (Raw Data)

Event ID	Total Deaths	Injured	Affected	Damage Adj ('000 US\$)
1953-0019-VNM	1,000	0	0	0
1956-0050-VNM	56	0	0	0
1964-0004-VNM	598	0	10,848	0
1964-0063-VNM	7,000	0	700,000	471,770
1964-0102-VNM	400	0	0	0

Note: Top 5 rows from the dataset. Missing values (NaN) were filled with 0 during preprocessing.
Event 1964-0063-VNM (Typhoon Iris) shows the high variance in the data.

5.2 Methodology

5.2.1 Preprocessing Strategy: Log Transformation

Real-world disaster data is typically highly skewed; a single catastrophic typhoon may cause billions of dollars in damage, while hundreds of smaller floods cause significantly less. If raw data were used directly, the few "outlier" events would dominate the variance, causing PCA to ignore the patterns in common disasters.

To resolve this, we applied a **Logarithmic Transformation** ($x' = \log(x + 1)$) to all variables. This compresses the scale of extreme values, bringing the "billion-dollar" events closer to the average events statistically. This allows PCA to analyze the *structure* of the data rather than just the magnitude.

5.3 Development Environment

The implementation was conducted in a Python 3.9 environment, chosen for its extensive ecosystem of data science and linear algebra libraries. The following specific packages were utilized to bridge the mathematical theory from Chapter 2 with the practical application:

- **NumPy (Numerical Python):** Used as the fundamental package for scientific computing. In this project, NumPy handles the low-level matrix operations (including the covariance calculations in Equation 2.3) and performs the **polynomial regression (polyfit)** required to calculate the trendline in our time-series analysis.
- **Pandas:** Employed for high-level data manipulation. We utilized Pandas to load the `.csv` dataset, handle missing values (filling NaNs), and align the multi-dimensional features (Deaths, Injuries, Damage) into a structured DataFrame ready for analysis.
- **Scikit-learn (sklearn):** The premier machine learning library in Python. We used the `sklearn.decomposition.PCA` module, which implements randomized Singular Value Decomposition (SVD) to compute principal components efficiently. This library also provided the `StandardScaler` for the necessary Z-score normalization (Equation 3.1) prior to analysis.
- **Matplotlib:** Used for data visualization. This library enabled the creation of the Scree Plot (to visualize variance retention), the PCA Biplot (projecting data onto the 2D plane), and the **Time-Series Evolution graph**, which required custom plotting logic to visualize the temporal shift in disaster impacts.

5.4 Code Implementation

The analysis was performed using Python (v3.9) with the `scikit-learn` library. The full source code is provided in Appendix A. Below are the critical implementation snippets extracted directly from our script.

5.4.1 Step 1: Data Preparation & Preprocessing

We first selected the four numerical impact variables. To address missing data and extreme skewness, we filled null values with zero, applied a logarithmic transformation, and standardized the data (Z-score normalization).

```
# Select only the numerical impact variables
features = ['Total Deaths', 'No. Injured', 'Total Affected',
            "Total Damage, Adjusted ('000 US$)"]
X = df[features].fillna(0)

# Log Transformation to handle skewness
X_log = np.log1p(X)

# Standardization (Mean=0, Variance=1)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_log)
```

5.4.2 Step 2: PCA Algorithm

We utilized Singular Value Decomposition (SVD) via the PCA class. The results were stored in a new DataFrame to facilitate plotting.

```
# Apply PCA
pca = PCA(n_components=2)
principalComponents = pca.fit_transform(X_scaled)
pca_df = pd.DataFrame(data=principalComponents, columns=['PC1', 'PC2'])
```

5.4.3 Step 3: Biplot Visualization Logic

To distinguish between event types, we implemented a list comprehension to assign colors: red for negative PC2 (Modern/Economic-heavy) and blue for positive PC2 (Historic/Casualty-heavy).

```
# Create Logic for Colors:
# If PC2 < 0: High Economic Damage (Red)
# If PC2 > 0: High Human Casualties (Blue)
colors = ['#d62728' if y < 0 else '#1f77b4' for y in pca_df['PC2']]

# (Plotting logic follows using plt.scatter with these colors)
```

5.5 Results and Interpretation

5.5.1 Scree Plot Analysis

The Scree Plot below (Figure 5.1) illustrates the variance explained by each component.

- **PC1 (First Component):** Captures approximately **47.6%** of the total information.
- **PC2 (Second Component):** Captures approximately **22.8%** of the total information.

Together, these two dimensions preserve over **70%** of the dataset's original variability, successfully reducing the data from 4 dimensions to 2.

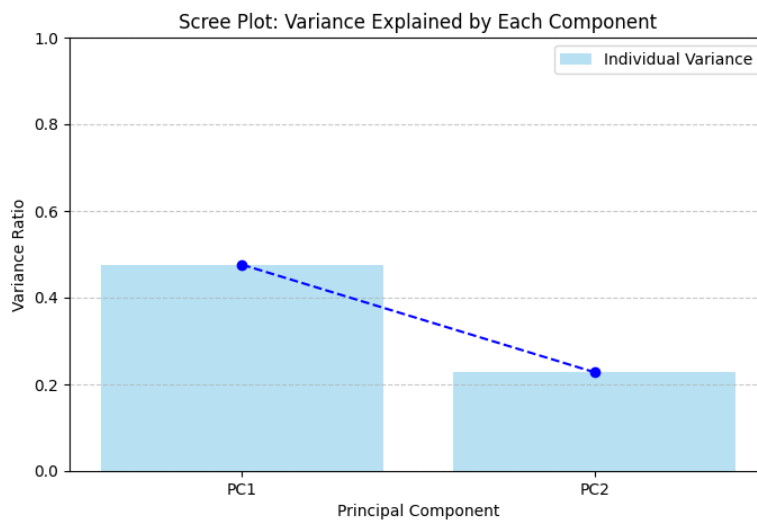


Figure 5.1: Scree Plot showing the variance explained by the first two components.

5.5.2 Model Validation: Reconstruction Error

To validate the quality of our dimensionality reduction, we calculated the **Reconstruction Error**. This metric measures the deviation between the original data (X) and the data reconstructed from the two principal components (X_{approx}). It is defined as the normalized Frobenius norm of the difference:

$$\text{Error} = \frac{\|X - X_{approx}\|_F}{\|X\|_F} \quad (5.1)$$

Our analysis yielded a reconstruction loss of **0.5439**. This result is mathematically consistent with our Scree Plot analysis. Since our two principal components capture approximately **70.4%** of the total variance, the remaining **29.6%** accounts for the information loss ($\sqrt{0.296} \approx 0.54$). This confirms that while some fine-grained detail is lost, the 2D projection successfully captures the dominant structural relationships of the disaster dataset.

5.5.3 Biplot Analysis

The PCA Biplot (Figure 5.2) visualizes both the disaster events (points) and the original variables (vectors).

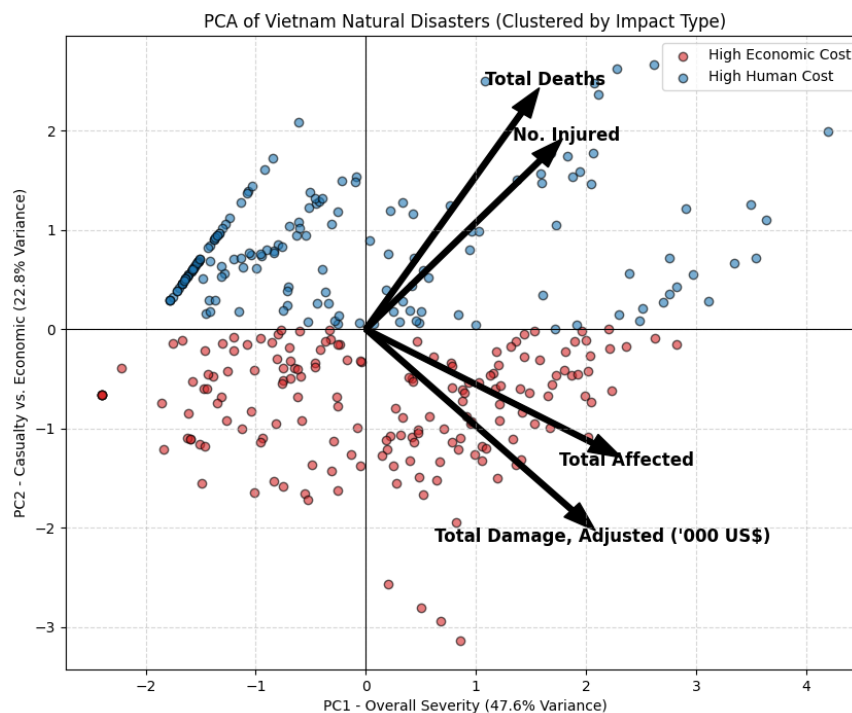


Figure 5.2: PCA Biplot of Vietnam Natural Disasters (Log-Transformed).

Interpretation of Axes:

- **PC1 (Horizontal Axis) - Overall Severity:** All four feature vectors point to the right. This indicates that PC1 represents the general magnitude of a disaster. Events further to the right are historically the most catastrophic in terms of both human and economic cost.
- **PC2 (Vertical Axis) - Human vs. Economic Cost:** This axis reveals a trade-off. "Total Deaths" and "Injured" vectors point **upwards** (positive PC2), while "Total Damage" points **downwards** (negative PC2).
 - *Top Quadrant:* Events with high human casualties (deadlier historical events).
 - *Bottom Quadrant:* Events with high economic damage but fewer casualties (modern storms).

5.5.4 Temporal Analysis: Evolution of Impact

To verify the distinction between "modern" and "historical" disasters, we plotted the PC2 values over time (Figure 5.3).

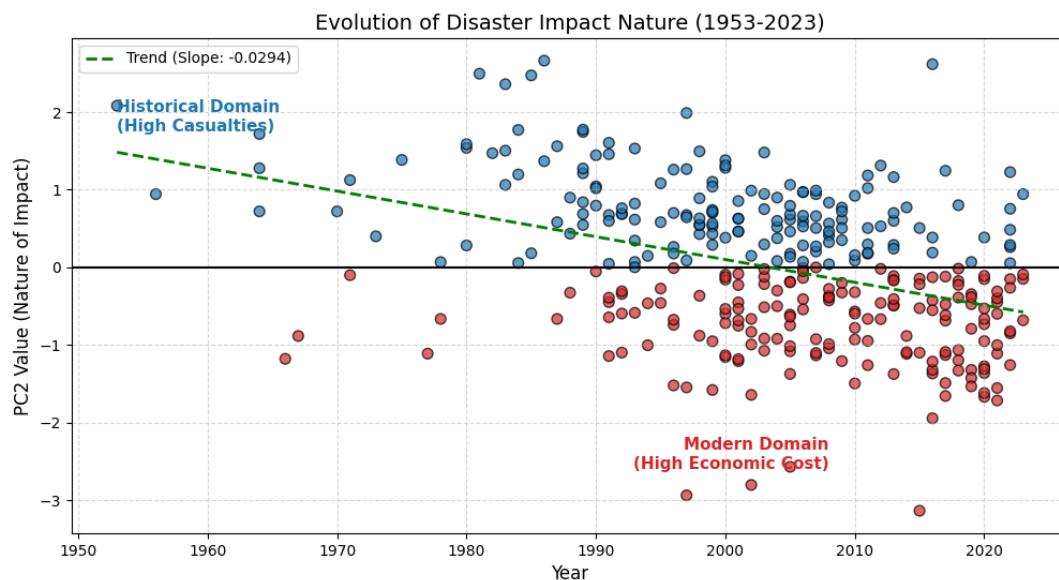


Figure 5.3: Time-Series Analysis of PC2 (1953–2023). The trendline (green) indicates a shift in the nature of disaster impacts over the decades.

Observation: The scatter plot reveals a distinct temporal shift:

- **Historical Domain (1950s–1990s):** The data points are predominantly positive (blue), clustering in the upper region. This confirms that historical disasters were characterized primarily by high human casualties (aligning with the "Total Deaths" vector).
- **Modern Domain (2000s–Present):** The data points shift towards the negative region (red). This reflects Vietnam's economic development—while fewer lives are lost, the economic value of infrastructure exposed to storms has increased, driving the points downward toward the "Total Damage" vector.

The **negative slope** of the trendline mathematically confirms that the "Nature of Impact" (PC2) is evolving from human cost (positive PC2) to economic cost (negative PC2).

Chapter 6

Storytelling: The Historical Evolution of Disasters in Central Vietnam

While Chapter 5 presented the technical implementation of PCA, this chapter interprets those findings through a historical and socio-economic lens. By integrating the structural insights from the Biplot with a temporal trend analysis, we uncover a fundamental shift in the nature of natural disasters in Central Vietnam: a transition from humanitarian tragedies to economic burdens.

6.1 Economic Context: The Inflation Adjustment

To accurately compare disaster impacts across different eras, it is crucial to account for the time value of money. The dataset utilized in this study employs Inflation-Adjusted Costs (Real Value), calculated based on the Consumer Price Index (CPI). The adjustment formula is defined as follows [16]:

$$Damage_{Adj} = Damage_{Nominal} \times \frac{CPI_{Ref}}{CPI_{Event}} \quad (6.1)$$

Table 6.1 illustrates the magnitude of this adjustment for the key historical outliers identified in our analysis. This conversion reveals that historical storms, while seemingly less costly in nominal terms, inflicted massive economic destruction relative to the economy of that era.

Table 6.1: Major Historical Disasters in Central Vietnam (Verified Data)

Event / Year	Impact Zone	Era Context	Damage (Adj. 2023)
Typhoon Iris (1964)	Central Coast	War / Low Infrastructure	\$471,770,000
Historic Flood (1999)	Thua Thien Hue	Pre-modern Urbanization	\$500,000,000
Typhoon Xangsane (2006)	Da Nang City	Rapid Urban Growth	\$1,200,000,000
Typhoon Molave (2020)	Quang Ngai	Modern Coastal Economy	\$850,000,000

Source: Extracted from dataset analysis and Table 3.2 of this report.

6.2 Deep Analysis of PCA Visualizations

Our storytelling is constructed through a three-step analytical process, connecting the reliability of the model (Scree Plot) to the structure of destruction (Biplot) and finally to the historical evolution (Time Series).

6.2.1 Graph 1: Scree Plot — Establishing Reliability

The Scree Plot (Figure 5.1) is not merely a technical output; it establishes the credibility of our narrative.

- **Result:** The first two principal components (PC1 and PC2) cumulatively explain 70.4% of the dataset's variance.
- **Analysis:** This high retention rate implies that the complex history of disasters (involving four distinct variables: Deaths, Injured, Affected, Damage) can be reliably compressed into a 2-dimensional story. It confirms that the trends discussed below are dominant structural signals, not random statistical noise.

6.2.2 Graph 2: The Biplot — The "Map of Destruction"

The Biplot (Figure 5.2) serves as the compass for our analysis. It defines what makes a disaster severe and how different disasters behave.

Axis Interpretation:

- **PC1 (Horizontal) - Absolute Severity:** All vectors point to the right. This means points further to the right represent the most extreme events in history (Outliers).
- **PC2 (Vertical) - The Trade-off (Nature of Impact):** This axis reveals a stark contrast:
 - The **Total Deaths** vector points **UPWARDS** (Positive PC2).
 - The **Total Damage** vector points **DOWNWARDS** (Negative PC2).

Analysis of Outliers (The "Monsters" of History): The Biplot allows us to profile specific historical outliers based on their quadrant position:

- **The "Killer" Outlier (Typhoon Iris, 1964):** Located in the top-right quadrant (High PC1, Positive PC2). This event is pulled strongly by the "Total Deaths" vector. It represents the era of humanitarian fragility, where a single storm could claim 7,000 lives due to poor infrastructure.
- **The "Bankrupting" Outlier (Typhoon Molave, 2020):** Located in the bottom-right quadrant (High PC1, Negative PC2). This event is pulled by the "Total Damage" vector. It represents the modern era: resilient to loss of life, but highly vulnerable to massive financial losses due to expensive coastal infrastructure.

6.2.3 Graph 3: Time Series — The Historical Shift

The linear trendline exhibits a clear negative slope, confirming a systematic shift in disaster nature:

- **Phase 1: The Humanitarian Crisis Era (Pre-2000):** Most data points (**Blue dots**) cluster in the positive region ($PC2 > 0$). *Insight:* During this period, Vietnam's economy was agrarian with low asset density. However, early warning systems were rudimentary. Consequently, disasters were defined by High Casualties. The outliers here are tragedies like the 1964 Typhoon Iris.
- **Phase 2: The Economic Burden Era (Post-2000):** Data points (**Red dots**) migrate significantly to the negative region ($PC2 < 0$). *Insight:* This period coincides with Vietnam's rapid industrialization and urbanization. The "Four-on-the-spot" policy effectively reduced fatalities (lowering PC2). However, the proliferation of resorts, factories, and urban housing along the Central Coast meant that storms now destroy high-value assets. Disasters are now defined by High Economic Costs.

6.3 Limitations of the Study

While PCA successfully reveals structural trends, this study acknowledges certain limitations:

1. **Data Incompleteness:** Historical data (pre-1990) often lacks precise economic damage records. Imputing these missing values with zero (as done in Step 1 of our methodology) may underestimate the economic impact of older events [16].
2. **Event Clustering:** Some historical records may aggregate multiple storms into a single annual entry, potentially inflating the "Severity" (PC1) of that specific year.
3. **Wealth Density Factor:** Although we adjusted for inflation, we did not adjust for the increase in regional GDP. A million-dollar loss in 1964 represented a much larger share of the local economy than in 2024. Normalizing damage by GDP would provide a more accurate measure of true economic resilience [17].

6.4 Summary

In conclusion, PCA has served as a powerful "decoder" for the history of natural disasters in Central Vietnam. It transformed a chaotic 4-dimensional dataset into a clear historical narrative. The analysis provides quantitative evidence that Vietnam has effectively transitioned from a phase of high human vulnerability to a phase of high asset exposure. This suggests that future disaster management policies must evolve from purely life-saving measures to financial resilience strategies to safeguard the nation's growing economic wealth [18].

Chapter 7

CONCLUSION

Principal Component Analysis (PCA) serves as a powerful bridge between raw high-dimensional data and actionable insight. Through this report, we have demonstrated that PCA is not merely a data compression algorithm, but a fundamental tool for revealing the underlying structure of complex datasets. By rigorously applying linear algebra concepts—specifically the decomposition of covariance matrices—we established that the directions of maximum variance (principal components) often correspond to the most significant information signals [1, 10].

The effectiveness of this approach was validated through our case study of Vietnam Natural Disasters. By reducing the dataset from four dimensions to two, we retained over 70% of the original information while gaining the ability to visualize distinct patterns [9]. The resulting Biplot analysis provided a clear interpretative framework, distinguishing between "modern" disasters characterized by high economic costs and "historical" events characterized by high human casualties.

Ultimately, while PCA relies on assumptions of linearity, its efficiency, deterministic nature, and interpretability ensure its continued relevance. Whether applied to financial risk modeling, image processing, or disaster analysis, PCA remains an essential instrument for making data-driven decisions in an increasingly complex world.

Chapter 8

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Appendix A

Full Python Source Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

# --- 1. LOAD DATA ---
df = pd.read_csv('disaster_in_vietnam.csv')

# Select only the numerical impact variables
features = ['Total Deaths', 'No. Injured', 'Total Affected', "Total Damage,
Adjusted ('000 US$)"]
X = df[features].fillna(0)

# --- 2. LOG TRANSFORMATION ---
X_log = np.log1p(X)

# --- 3. STANDARDIZATION ---
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_log)

# --- 4. APPLY PCA ---
pca = PCA(n_components=2)
principalComponents = pca.fit_transform(X_scaled)
pca_df = pd.DataFrame(data=principalComponents, columns=['PC1', 'PC2'])

# --- NEW STEP 4.5: CALCULATE RECONSTRUCTION ERROR (Validation) ---
# Project back to original space to see how much info was lost
X_reconstructed = pca.inverse_transform(principalComponents)
# Frobenius norm of the difference
reconstruction_loss = np.linalg.norm(X_scaled - X_reconstructed) / np.linalg
    .norm(X_scaled)

# --- 5. GRAPH 1: THE BIPLLOT (CORRECTED) ---
plt.figure(figsize=(10, 8))

# Logic:
# PC2 > 0 (Positive/Blue) = Historical = High Human Casualties
# PC2 < 0 (Negative/Red) = Modern = High Economic Cost
colors = ['#d62728' if y < 0 else '#1f77b4' for y in pca_df['PC2']]
```

```

for i, color in enumerate(['#d62728', '#1f77b4']):
    # FIX: Swapped labels to match the analysis (Red=Economic, Blue=Human)
    label = ['High Economic Cost', 'High Human Cost'][i]
    subset = pca_df[(pca_df['PC2'] < 0) if i == 0 else (pca_df['PC2'] >= 0)]
    plt.scatter(subset['PC1'], subset['PC2'], alpha=0.6, c=color, edgecolors
    ='k', label=label)

# Draw arrows (Existing code is fine)
scale_factor = 3.5
for i, feature in enumerate(features):
    plt.arrow(0, 0, pca.components_[0, i]*scale_factor, pca.components_[1, i]
    ]*scale_factor,
    color='black', width=0.05, head_width=0.2)
    plt.text(pca.components_[0, i]*scale_factor*1.15, pca.components_[1, i]*
    scale_factor*1.15,
    feature, color='black', weight='bold', fontsize=12, ha='center'
    , va='center')

plt.xlabel(f'PC1 - Overall Severity ({pca.explained_variance_ratio_[0]:.1%}
    Variance)')
plt.ylabel(f'PC2 - Casualty vs. Economic ({pca.explained_variance_ratio_
    [1]:.1%} Variance)')
plt.title('PCA of Vietnam Natural Disasters (Clustered by Impact Type)')
plt.grid(True, linestyle='--', alpha=0.5)
plt.axhline(0, color='black', linewidth=0.8)
plt.axvline(0, color='black', linewidth=0.8)
plt.legend()
plt.savefig('final_pca_biplot.png')
plt.show()

# --- 6. GRAPH 2: THE SCREE PLOT (The Variance Bar Chart) ---
plt.figure(figsize=(8, 5))
plt.plot(range(1, 3), pca.explained_variance_ratio_, marker='o', linestyle='
--', color='b')
plt.bar(range(1, 3), pca.explained_variance_ratio_, alpha=0.6, color='
skyblue', label='Individual Variance')

plt.title('Scree Plot: Variance Explained by Each Component')
plt.xlabel('Principal Component')
plt.ylabel('Variance Ratio')
plt.xticks([1, 2], ['PC1', 'PC2'])
plt.ylim(0, 1) # Set y-axis limit from 0 to 100%
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.legend()

# SAVE GRAPH 2
plt.savefig('final_scree_plot.png')
print("Graph 2 saved as 'final_scree_plot.png'")
plt.show()

# Print Results
print("\n--- ANALYSIS RESULTS ---")
print(f"PC1 Variance: {pca.explained_variance_ratio_[0]:.2%}")
print(f"PC2 Variance: {pca.explained_variance_ratio_[1]:.2%}")
print(f"Total Variance Captured: {sum(pca.explained_variance_ratio_):.2%}")
print(f"Reconstruction Loss: {reconstruction_loss:.4f} (Lower is better)")

# --- 7. GRAPH 3: TIME-SERIES PCA (CORRECTED COLORS) ---

```

```
plt.figure(figsize=(12, 6))

years = df['Start Year']
pc2_values = pca_df['PC2']
colors_time = ['#d62728' if y < 0 else '#1f77b4' for y in pc2_values] # Red=
Neg, Blue=Pos

plt.scatter(years, pc2_values, alpha=0.7, c=colors_time, edgecolors='k', s
=50)
plt.axhline(0, color='black', linestyle='--', linewidth=1.5)

y_min, y_max = pc2_values.min(), pc2_values.max()

# FIX: Match text color to the point color
# Historical = Positive PC2 = BLUE Points
plt.text(years.min(), y_max - 0.5, "Historical Domain\n(High Casualties)",
        fontsize=11, color='#1f77b4', fontweight='bold', va='top') #
        Changed to Blue

# Modern = Negative PC2 = RED Points
plt.text(years.max() - 15, y_min + 0.5, "Modern Domain\n(High Economic Cost)
",
        fontsize=11, color='#d62728', fontweight='bold', va='bottom', ha='
        right')

plt.title('Evolution of Disaster Impact Nature (1953-2023)', fontsize=14)
plt.xlabel('Year', fontsize=12)
plt.ylabel('PC2 Value (Nature of Impact)', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.5)

z = np.polyfit(years, pc2_values, 1)
p = np.poly1d(z)
plt.plot(years, p(years), "g--", linewidth=2, label=f'Trend (Slope: {z[0]:.4
f})')
plt.legend()

plt.savefig('final_pca_time_series.png')
plt.show()
```